

BRAC UNIVERSITY

Department of Computer Science and Engineering

SET: B

Course: CSE423

Examination: Quiz 2

Semester: Spring 2025

Duration: 20 min

Marks: 10

Name:

ID:

1. A rectangular viewport is defined with its top-right corner at (40, 80). The viewport has a width that is twice its height. The bottom-left corner of the viewport lies on the x-axis. A line segment has the startPoint A (-20, -10) and endpoint B (50, 90). Apply the **Cohen-Sutherland Line Clipping Algorithm** and identify whether the line is "partially inside", "completely inside", or "completely outside". If it is the first case, run the algorithm to calculate new endpoints for the line segment so that it is inside the clipping window. **[10]**
2. **[Bonus]** Why does the Cyrus Beck Line Clipping Algorithm not work with a concave polygon clip region? Give proper reasoning.